

# Ariba Shakeel

PhD Researcher in Applied Mathematics

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## Research Interests

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**Current:** Mathematical Biology, Mathematical Modelling, Applied Differential Equations

**Previous:** Fluid Dynamics, Boundary-Layer Theory, Rotating Flow, Computational Fluid Mechanics

## Education

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**University of Birmingham**

Birmingham, United Kingdom

*Doctor of Philosophy (PhD) in Applied Mathematics*

*September 2024–Present*

**Supervisors:** Dr Meurig Thomas Gallagher and Prof. David Dayson Smith

**Project Title:** “*Mathematical Modelling to Understand the Treatment of Adrenal Insufficiency*”

My research develops a mathematical modelling toolkit to investigate cortisol replacement therapy in adrenal insufficiency. The models incorporate key physiological mechanisms, including cortisol–protein binding, cortisol–cortisone interconversion, tissue distribution and cortisol excretion. The aim is to explore different dosing strategies and understand the physiological mechanisms underlying cortisol dynamics and inter-patient variability.

**Methods and approaches include:**

- Development and analysis of mechanistic ordinary differential equation (ODE) models;
- Asymptotic and perturbation analysis of multi-timescale physiological processes;
- Numerical simulation, parameter estimation and optimisation using clinical data;
- Structural and practical identifiability analysis to assess model parameters and predictions.

**National University of Sciences and Technology (NUST)**

Islamabad, Pakistan

*Master of Science in Mathematics*

*September 2021–August 2023*

**Supervisor:** Prof. Meraj Mustafa Hashmi

**CGPA:** 3.55/4.00

**Research Interests:** Fluid Dynamics, Boundary-Layer Theory, Rotating Flow, Computational Fluid Mechanics

**Dissertation Title:** “*Investigation of rotating flows over a stationary plane surface*”

**University of Sargodha (UOS)**

Sargodha, Pakistan

*Bachelor of Science in Mathematics*

**CGPA:** 3.36/4.00

## Previous Research Experience

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**National University of Sciences and Technology (NUST)**

Islamabad, Pakistan

*Research Assistant*

*March 2023–August 2023*

**Supervisor:** Prof. Meraj Mustafa Hashmi

**Project Title:** “*Modelling of MHD fluid flow over-stretching wedge using variable fluid properties*”

- Investigated MHD boundary-layer flow with variable fluid properties using mathematical and numerical methods.
- Applied MATLAB’s `bvp4c` solver to obtain numerical solutions of the resulting boundary-value problem.

## Publications

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1. **A. Shakeel** and M. Mustafa, “Assessment of Bödewadt flow over a stretchable porous surface with variable physical properties: A comparative study,” *Numerical Heat Transfer, Part B: Fundamentals*,

pp. 1–14, 2023. DOI: 10.1080/10407790.2023.2274452.

2. **A. Shakeel**, M. Mustafa, and K. Naganthran, “A numerical exploration of boundary layer flow and heat beneath a generalized vortex flow by considering variable fluid properties,” p. 2550136, 2025. DOI: 10.1142/S0217984925501362.

## Conference and Seminar Presentations

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### Oral Presentations

- 2026** “Mathematical Modelling to Understand the Treatment of Adrenal Insufficiency,” Applied Mathematics Postgraduate Conference, University of Warwick, Coventry, UK.
- 2026** 3-minute thesis talk about “Mathematical Modelling to Understand the Treatment of Adrenal Insufficiency” at British Early career Mathematician Colloquium, University of Birmingham, UK.
- 2026** “Mathematical Modelling to Understand the Treatment of Adrenal Insufficiency: Exploring Continuous Intravenous Infusion” Lunchtime Applied Mathematics Seminar (LAMS), University of Birmingham, UK.
- 2025** “Mathematical Modelling: Cortisol infusion protocols to understand the treatment of Adrenal insufficiency” British Early career Mathematician Colloquium, University of Birmingham, UK.
- 2025** “Numerical and Asymptotic Analysis of Cortisol-Protein Binding Model: Exploring saturating case” Mathematical and Computational Biology Seminar Series, University of Birmingham, UK.
- 2025** “Numerical and Asymptotic Analysis of Cortisol-Protein Binding Model: Exploring Non-saturating Dynamics” Lunchtime Applied Mathematics Seminar (LAMS), University of Birmingham, UK.

### Poster Presentations

- 2025** “Mathematical Modelling: Cortisol infusion protocols to understand the treatment of Adrenal insufficiency” in Women and Non-binary Doctoral Researchers in STEM (WoNDRS) Conference at University of Liverpool, Liverpool, UK.
- 2025** “Numerical and Asymptotic Analysis of Cortisol infusion protocols for the treatment of Adrenal Insufficiency”, in UK Mathematical Biology Conference, University of Birmingham, UK.
- 2025** “Mathematical Modelling for the treatment of Adrenal Insufficiency” in Celebrating WOMEN in MATHEMATICAL BIOLOGY DAY at University of Oxford, UK.

## Teaching Experience

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### University of Birmingham

*Postgraduate Teaching Assistant (PGTA)*

Birmingham, United Kingdom

*October 2024–Present*

- Demonstrate and support undergraduate and postgraduate mathematics classes.
- Mark assignments and examination scripts.
- Support computer-based mathematics sessions using MATLAB and L<sup>A</sup>T<sub>E</sub>X.
- Undertake examination invigilation.

## Awards and Scholarships

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- 2024–Present** **EPSRC Doctoral Studentship**, University of Birmingham, UK. Covers tuition fees and doctoral stipend.

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|------------------|---------------------------------------------------------------------------------------------------------------|
| <b>2017–2021</b> | <b>Higher Education Commission (HEC) Merit-Based Scholarship</b> , Pakistan. Awarded based on academic merit. |
| <b>2018</b>      | <b>Prime Minister’s Laptop Award</b> , Pakistan. Awarded for academic excellence.                             |
| <b>2015–2017</b> | <b>Merit-Based Full Tuition Fee Waiver</b> . Awarded for academic performance at intermediate level.          |

## Academic Service & Outreach

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| <b>2026</b> | <b>Co-organiser, Lunchtime Applied Mathematics Seminar (LAMS)</b> , School of Mathematics, University of Birmingham, UK.                                                                           |
| <b>2025</b> | <b>PhD Student Panellist, PhD Pathways</b> , University of Birmingham, UK. Participated as a speaker, sharing experiences and advice on PhD study, funding, research and academic career pathways. |
| <b>2025</b> | <b>Volunteer, PhDYourWay Event</b> , University of Birmingham, UK. Supported student Q&A, answering questions and providing guidance on PhD study, applications and doctoral pathways.             |

## Additional Volunteering

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| <b>2023</b> | Volunteer, World Mental Health Day, School of Natural Sciences, NUST, Pakistan.          |
| <b>2023</b> | Volunteer, National Disaster Resilience Day, School of Natural Sciences, NUST, Pakistan. |
| <b>2023</b> | Volunteer, Astronomy Workshop, School of Natural Sciences, NUST, Pakistan.               |

## Selected Conferences, Workshops & Training

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| <b>2026</b> | STEM Thesis Writing Sessions, University of Birmingham, UK.                    |
| <b>2026</b> | Workshop run by G-Research at AMP Conference, University of Warwick, UK.       |
| <b>2025</b> | Software Carpentry Git Training Course, University of Birmingham, UK.          |
| <b>2023</b> | International Workshop on Fluid Dynamics, Malaysia.                            |
| <b>2023</b> | Data Science Boot Camp, School of Natural Sciences, NUST, Islamabad, Pakistan. |
| <b>2022</b> | Applications of MATLAB in Science and Engineering, NUST, Islamabad, Pakistan.  |

## Technical & Research Skills

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| <b>Programming &amp; Scientific Computing</b> | MATLAB, Python                                                                               |
| <b>Mathematical Software</b>                  | Mathematica, L <sup>A</sup> T <sub>E</sub> X, XPPAUT                                         |
| <b>Mathematical Modelling</b>                 | Ordinary differential equations, mechanistic modelling, multi-timescale modelling            |
| <b>Analytical Methods</b>                     | Asymptotic analysis, perturbation methods, matched asymptotic analysis                       |
| <b>Computational Methods</b>                  | Numerical simulation, parameter estimation, optimisation, sensitivity analysis, data fitting |
| <b>Model Analysis</b>                         | Structural and practical identifiability analysis                                            |

## Languages

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| English | Full working proficiency |
| Urdu    | Native proficiency       |

## References

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Available upon request.